Lucas Krippendorff

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Education

University of Pennsylvania Aug 2024 – May 2028
B.S.E. Electrical Engineering *Philadelphia, PA*
- **Coursework:** Embedded Systems/Microcontroller Laboratory; Circuit-Level Modeling, Design, and Optimization for Digital Systems; Digital Integrated Circuits and VLSI-Fundamentals (graduate, in progress)

Experience

SubVysion (YC S26) Mar 2026 – Present
Hardware Engineer Intern · First Engineering Hire & Hardware Lead *Philadelphia, PA*
- Owned rover hardware end to end, both mechanical and electrical: turned the CEO's autonomous-scanning goal into rover designs and iterated with the software intern on fixes found in field testing
- Built the first autonomous GPR mapping rover in under two weeks, replacing a manual sensor cart limited to 0.5 m/s; later revisions scan at 3 m/s on pavement
- Upgraded compute from Jetson Xavier NX to Orin Nano to Orin NX as the 6× speed increase multiplied data from three ZED X stereo cameras (GMSL2), RTK GNSS, IMU, and GPR; now adding LiDAR (Livox Mid-360 vs. Unitree L2)
- Replaced skid-steer drive with casters after skidding made turns unpredictable and power-hungry; moved the casters to the front after studying a large agricultural rover, which proved more stable in testing
- Architected 36 V LiFePO4 power distribution with an anti-spark main switch, a 240 W buck converter feeding a fused 12 V bus, and brake resistors sized for ODrive regen
- Designed sensor mounts and electronics packaging in Onshape, using non-metallic G10/FR4 and ABS enclosures for GPR compatibility; wrote the BOM and visual assembly guide the whole company used to build a rover together in San Francisco

3D Chimera Jun 2025 – Jul 2025
Engineering Intern · 3D-printing solutions & Bambu Lab reseller *Miami, FL*
- Learned woodworking to build a shop storage shelf with 3D-printed dividers; repaired and calibrated Bambu Lab X1E and H2D printers and supported tensile, shear, and sealing tests of printed parts

Projects

USB-C Speaker & Discrete Loudness Meter | *Personal Project* Aug 2026 – Present
- Designed a USB-C amp board in KiCad that enumerates as a USB sound card (PCM2704), negotiates 12 V over PD (CH221K), and drives a TDA7297 class-AB stage; cut assembly cost over 4 JLCPCB revisions by moving to 0805 SMD and choosing parts JLCPCB or the Penn lab already stocks
- Built a 20-LED stereo loudness meter from discrete MMBT3904 stages as a flash-ADC-style ladder, sizing each base divider by hand so thresholds fall at equal dB steps, snapped to E24 values

22nm CMOS Datapath & Memory | *UPenn ESE 3700* Jan 2026 – May 2026
- Designed an 8-bit ripple-carry adder and a synchronous 16×4 6T SRAM at the transistor level (22nm HP PTM, 0.8 V)
- Verified SRAM reads and writes in SPICE up to 3.36 GHz (298 ps minimum period), 6.7× above the 500 MHz spec
- Cut worst-case carry delay 27% using a transmission-gate XOR2 over a NAND baseline, checked with Elmore and SPICE

Six-DOF Robotic Arm | *Personal Project* May 2025 – Aug 2025
- Designed and built a 0.8 m reach arm with custom 26:1 cycloidal and 5:1 planetary gearboxes with integrated bearings on NEMA-17 steppers
- Built an inverse-kinematics solver and trajectory planner in Python, streaming step pulses to C++ Arduino firmware, with a JavaScript/Meshcat interface for real-time waypoint control

Cat–Dog CNN Classifier | *UPenn ENGR 1050* Dec 2024
- Trained a PyTorch CNN on 24,000 images with CUDA; diagnosed overfitting from train/validation divergence (100% vs. 82%), deepened the network to raise validation accuracy to 90%, and reached 98% on a held-out test set

Activities, Skills & Interests

Activities: Penn Mars Rover Club, Penn Aerial Robotics Team, Penn Outdoors Club, Penn Astronomy Club
Programming: Python (PyTorch, CUDA), C/C++ (embedded firmware), JavaScript, Git
Hardware: NVIDIA Jetson, STM32 / ESP32 / ATmega, BLDC & stepper control (ODrive), CAN / UART / I2C / SPI, RTK GNSS, stereo cameras, GPR, LiFePO4 / LiPo systems, buck converters, PCB design, SPICE
Lab: Oscilloscope, multimeter, bench power supply, soldering and rework
Fabrication & CAD: Onshape, SolidWorks, Fusion 360, KiCad, Electric VLSI, 3D printing, G10/FR4 composites, wiring harnesses
Interests: music production (released as Hazed on Spotify and Apple Music), DJing, building custom PCs